

12. If $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$ then $A^{-1} = \underline{\hspace{2cm}}$.

(A) $\begin{bmatrix} -5 & 3 \\ -2 & -1 \end{bmatrix}$.

(B) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$.

(C) $\begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$

(D) None of the above.

13. Let A be a square matrix.

P : A^{-1} exist for every square matrix A. Q : There are square matrices A whose inverse does not exist.

(A) P is true and Q is false.

(B) P is false and Q is true.

(C) P and Q are false.

(D) P and Q are true.

14. The maximum value of the objective function $z = 3x + 4y$, subject to the following constraints

$$2x + y \leq 4$$

$$-x + 2y \leq 4$$

$$x \geq 0$$

$$y \geq 0$$

(A) 6.

(B) 12.

(C) 18.

(D) 24.

15. What is the solution of the following linear programming problem ?

$$\text{Minimize } Z = 2x + 4y$$

$$\text{subject to } x + 2y \geq 10$$

$$3x + y \geq 10$$

$$x \geq 0$$

$$y \geq 0.$$

(A) 5.

(B) 10.

(C) 15.

(D) 20.

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS-UG)

Mathematics

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A

*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Does the line $y = -x + 5$ intersect the point $(3, -1)$? Why ?
2. Let $g(x) = -4x + k$ where k is a constant. If $g(3) = 5$, find the value of k .
3. Solve the system of equations $3x + y = 5$, $3x = 6$.
4. Write the augmented matrix for the system of equations $3x + y = 6$, $2x + 5y = 15$.
5. Find values of x, y if $\begin{bmatrix} 3 & 4 \\ -8 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2y \\ x & 1 \end{bmatrix}$.
6. Find the product of matrices : $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} -1 & 5 \\ 0 & 3 \end{bmatrix}$.
7. Graph the linear inequality $x \leq 3y$.
8. Define the term *corner point*. State the corner point theorem.
9. Give an example for a maximization problem in standard form with 2 variables.
10. Sketch the feasible region for the linear programming problem :
Maximize $Z = 2x + 3y$ subject to $x \geq 0, y \geq 0$.
11. What are the conditions to be satisfied to call a linear programming problem to be in standard minimum form ?

Turn over

12. Write the matrix form of the linear programming problem :

$$\text{Minimize } W = 8y_1 + 16y_2$$

$$\begin{aligned} \text{subject to } & y_1 + 5y_2 \geq 9 \\ & 2y_1 + 2y_2 \geq 10 \\ & y_1 \geq 0, y_2 \geq 0. \end{aligned}$$

(8 × 3 = 24 marks)

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. In recent years, the percentage of the U.S. population age 18 and older who smoke has decreased at a roughly constant rate, from 24.1% in 1998 to 20.6% in 2008. Find the equation describing this linear relationship.

14. Solve the system of equations :

$$3x + 10y = 115$$

$$11x + 4y = 95$$

using echelon method.

15. A convenience store sells 23 sodas one summer afternoon in 12-, 16-, and 20-oz cups (small, medium, and large). The total volume of soda sold was 376 oz. Suppose that the prices for a small, medium, and large soda are \$1, \$1.25, and \$1.40, respectively, and that the total sales were \$28.45. How many of each size did the store sell ?

16. Solve the following system of equations using the inverse of the coefficient matrix :

$$x + 3y - 2z = 4$$

$$2x + 7y + 3z = 8$$

$$3x + 8y + 5z = -4.$$

17. Graph the feasible region for the following system of inequalities and tell if it is bounded or unbounded :

$$3x - 2y \geq 6$$

$$x + y \leq -5$$

$$y \leq -6$$

18. Find the maximum value of the objective function $z = 3x + 4y$, subject to the constraints :
 $-x + 2y \leq 4, x \geq 0, y \geq 0.$